SIS 3204Y(5) – Natural Language Processing

Week 1 – Revision Questions

Answer all questions. Marks are indicated in brackets.

Question 1 (2 marks)
Define Natural Language Processing (NLP).

Question 2 (4 marks)
State FOUR real-world applications of NLP.

Question 3 (6 marks)
Explain why human language is difficult for computers to understand.

Question 4 (6 marks)
Differentiate between Traditional NLP and Modern NLP.

Question 5 (6 marks)
Describe the NLP pipeline using a labelled diagram.

Question 6 (3 marks)
What is tokenisation? Give one suitable example.

Question 7 (4 marks)
Explain the role of word embeddings in Natural Language Processing.

Question 8 (7 marks)
Describe the evolution of NLP from rule-based systems to Large Language Models (LLMs).

Question 9 (5 marks)
What is a Transformer and why is it important in modern NLP?

Question 10 (4 marks)
What are Large Language Models (LLMs)? Give TWO examples.

Question 11 (5 marks)
Explain Retrieval-Augmented Generation (RAG) and discuss its purpose.

Question 12 (3 marks)
What is hallucination in a Large Language Model?

Question 13 (6 marks)
Discuss THREE ethical challenges associated with Natural Language Processing systems.

Question 14 (15 marks)

Explain how NLP has evolved from rule-based systems to modern Large Language Models (LLMs). Discuss the benefits and challenges of this evolution, using suitable examples where appropriate.